



Design of a Self-Assembly System of Three-dimensional Structures using Autonomous Construction Blocks

Kleber M. Cabral, Sidney N. Givigi Jr,
Sergio R. B. dos Santos, Peter T. Jardine

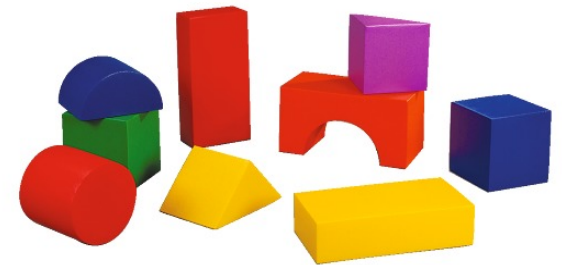
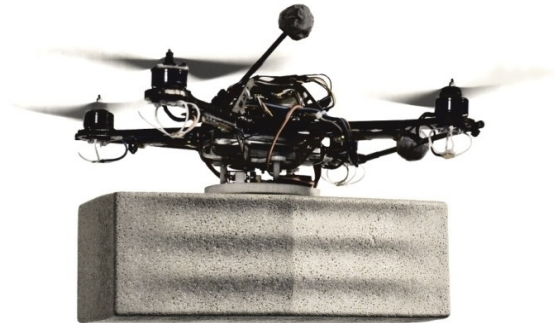
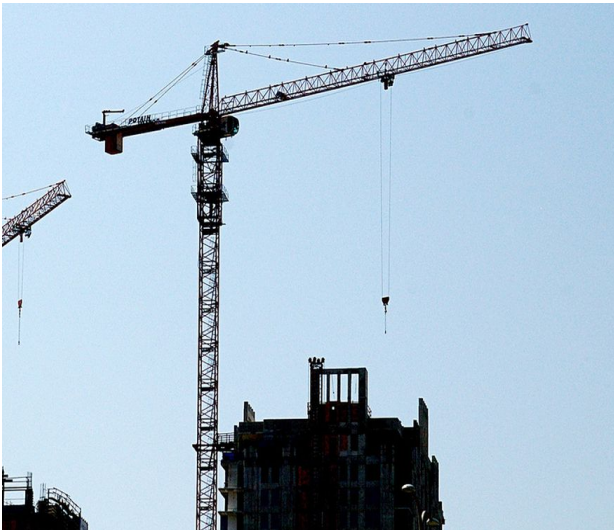


Agenda

- Introduction
 - Autonomous construction.
 - Actuated blocks
- High level architecture
 - General and multi-agent
- Simulated experiment
 - Pyramid and Stairs
- Results
- Conclusion

Autonomous construction

- Build a structure from parts
 - Manually or autonomously transported
 - Actuated (intelligent) parts

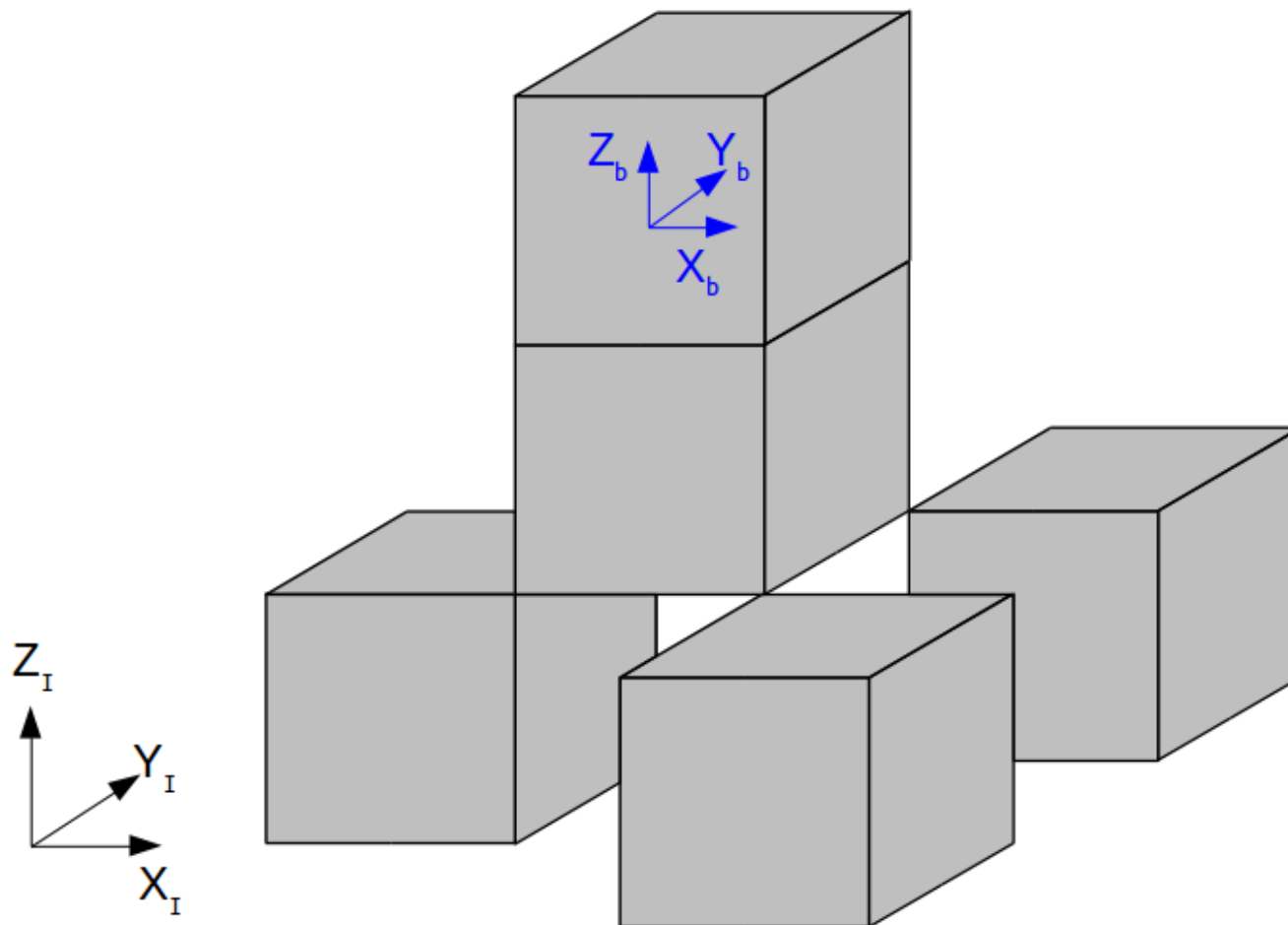




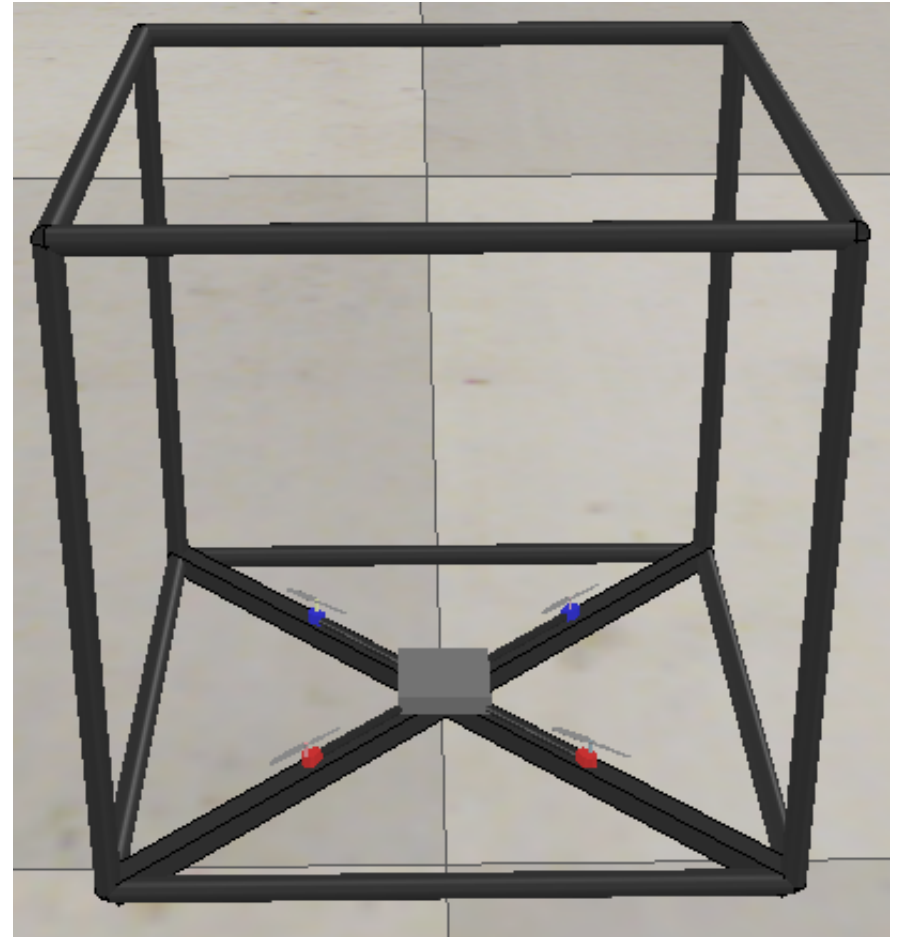
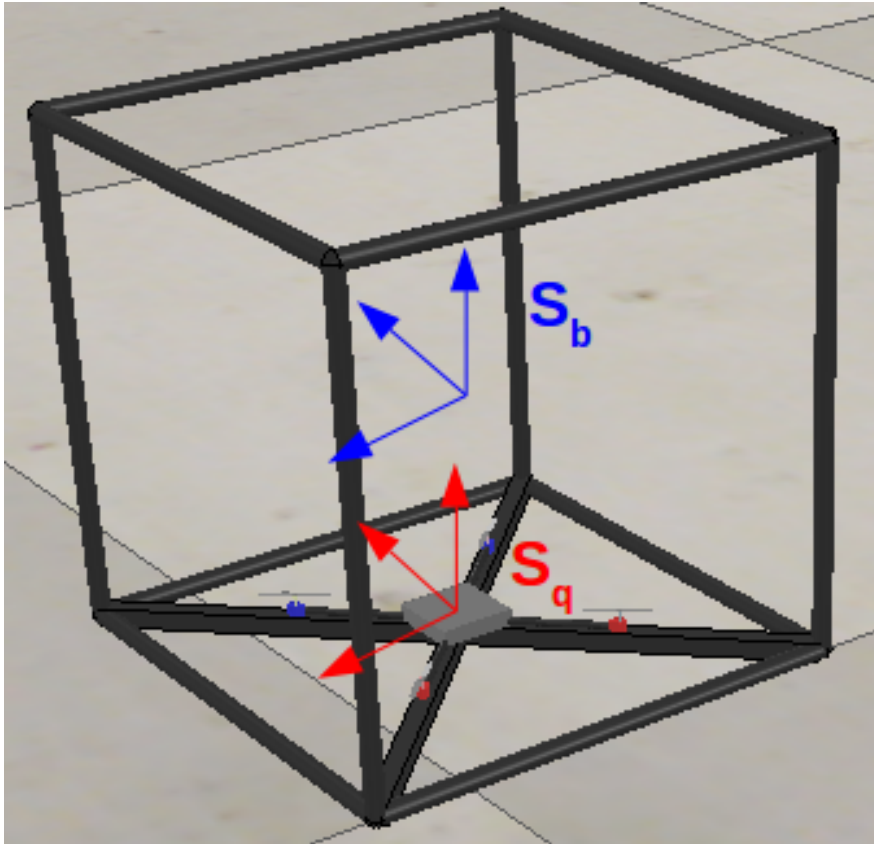
Objective

Propose a high level architecture for a autonomous construction where the vehicles place themselves as the building blocks of the structure.

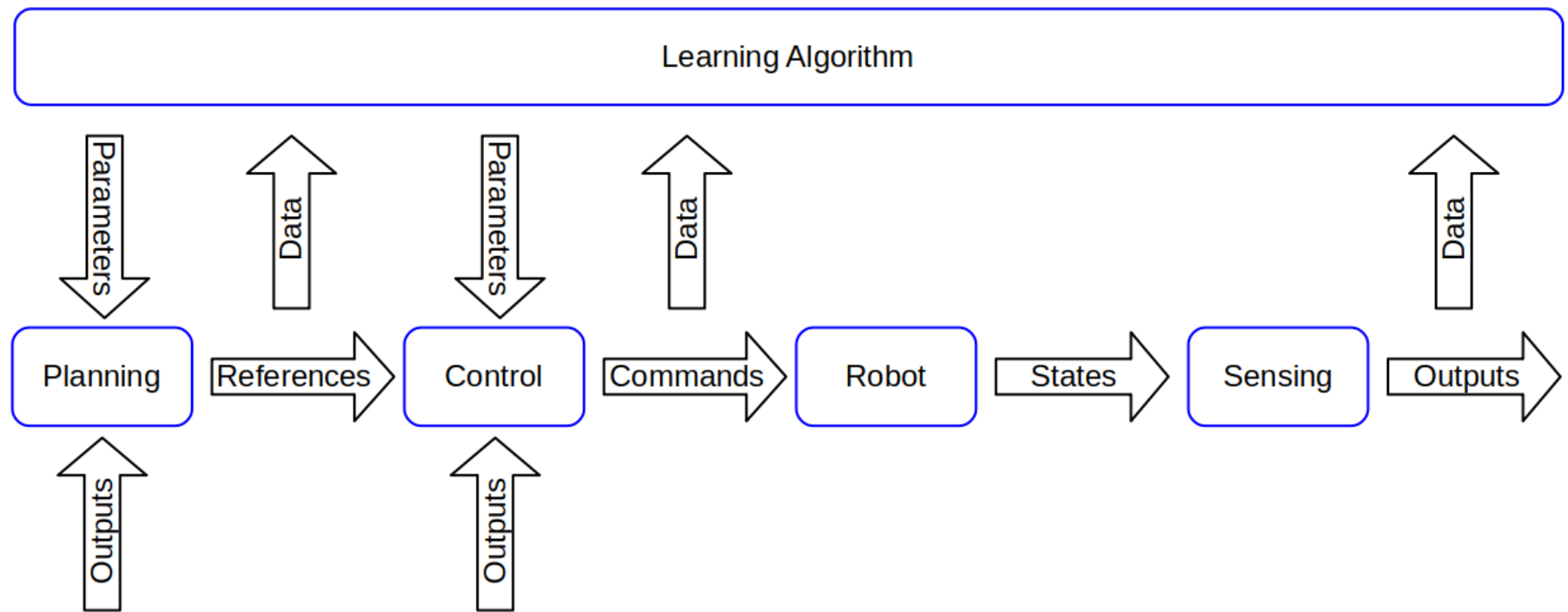
Autonomous Construction



Actuated Building Blocks



High-Level Architecture





High-Level Architecture

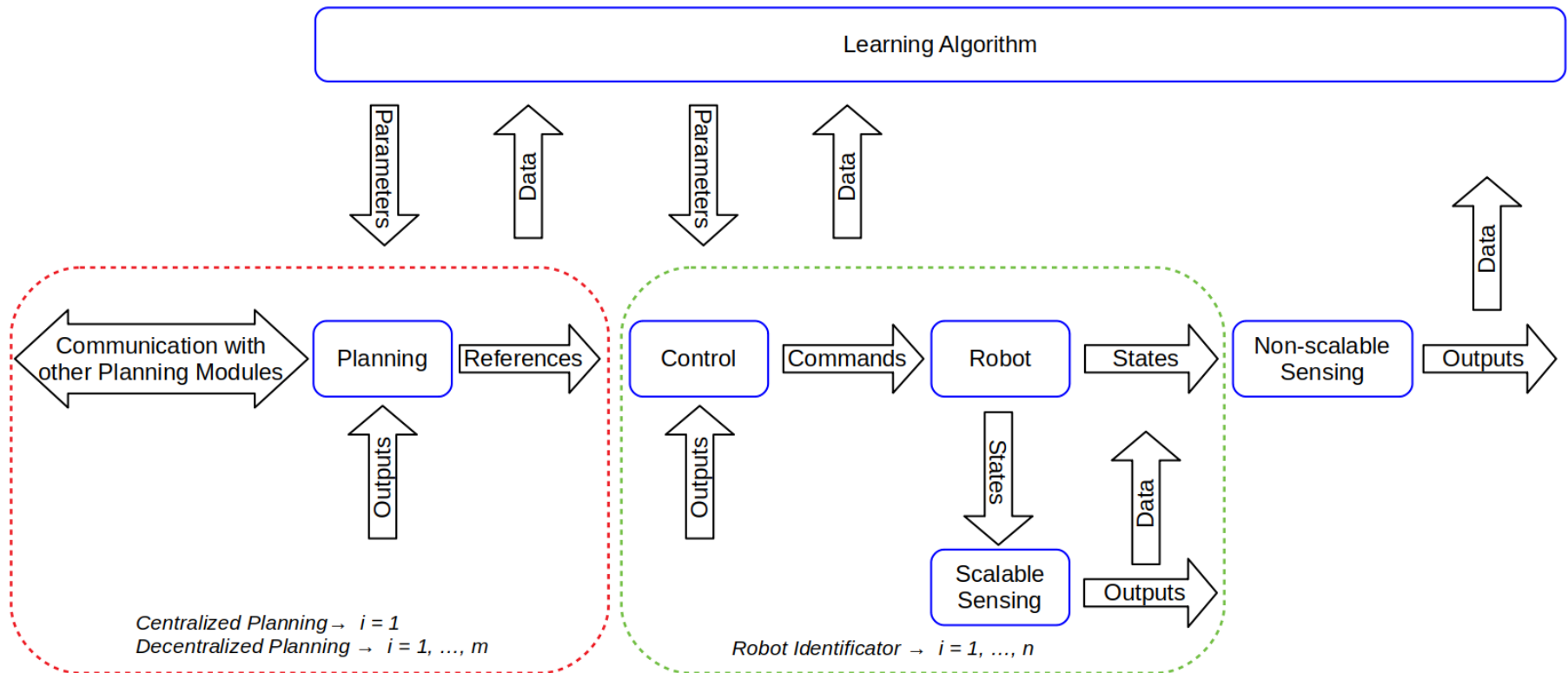
- System Modules
 - Planning, Control, Robot and Sensing
- System Information
 - Parameters, references, outputs, data, states
- Works for a single vehicle
- Shows the general idea



High-Level Architecture for Multi-robots

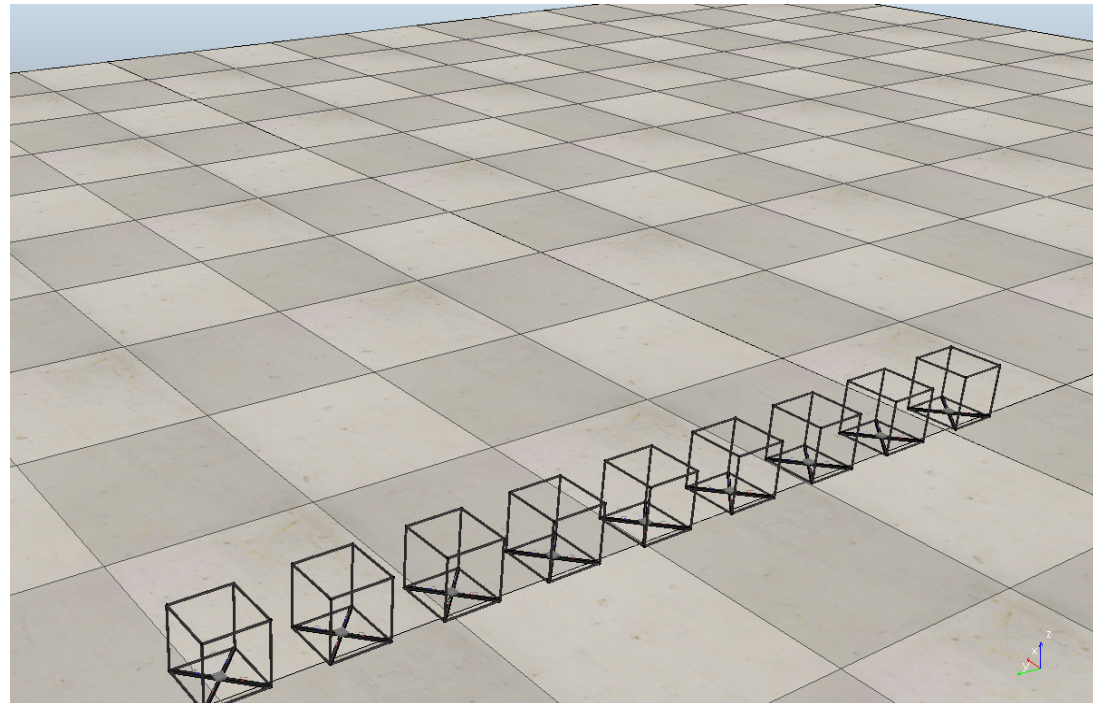
- The quantity of some elements may vary with the number of robots
 - Sensing
 - Non-scalable
 - Scalable
 - Planning algorithms
 - Centralized planning
 - Distributed planning

High-Level Architecture for Multi-robots

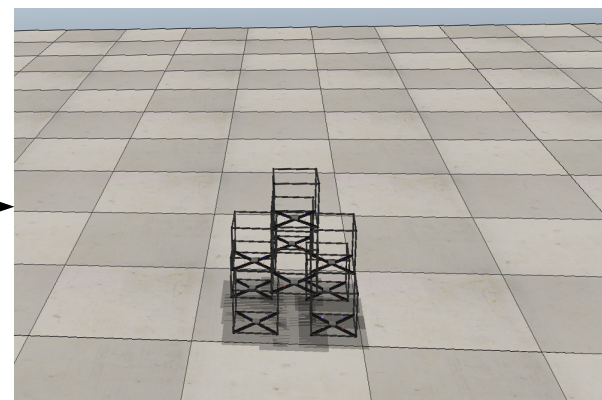
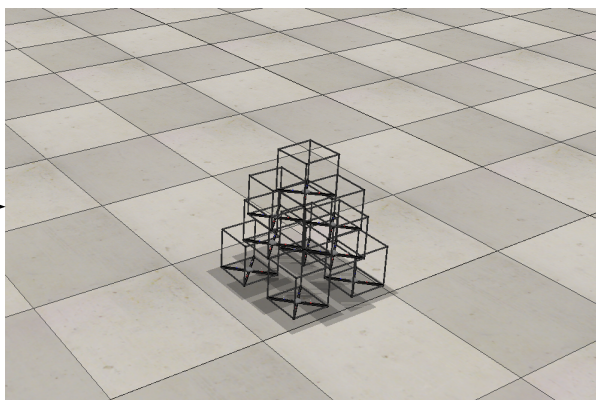
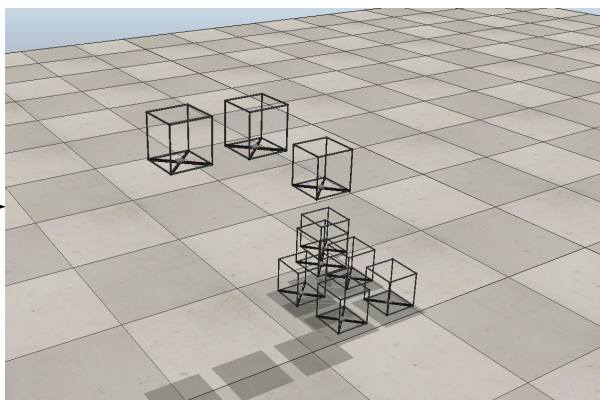
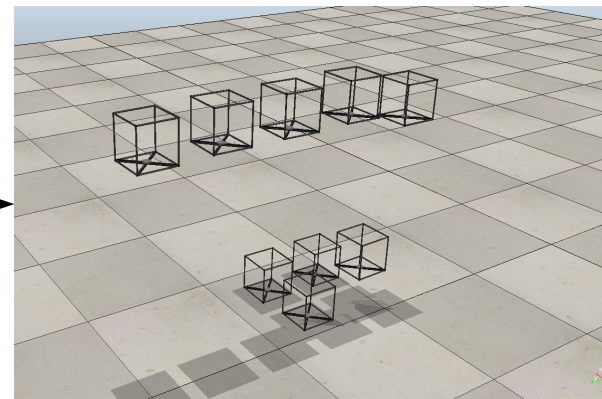
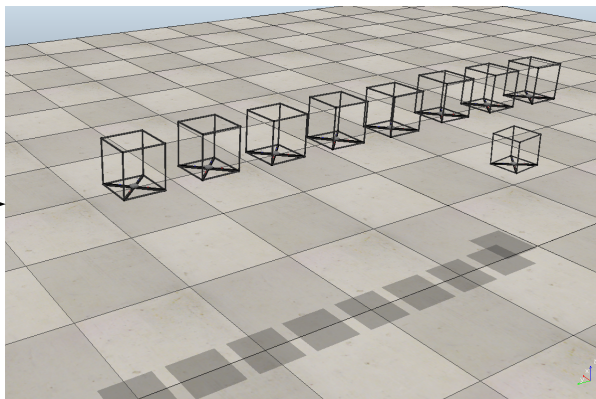
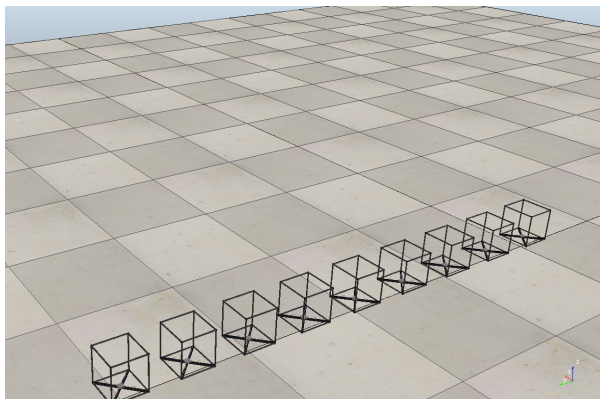


Simulation

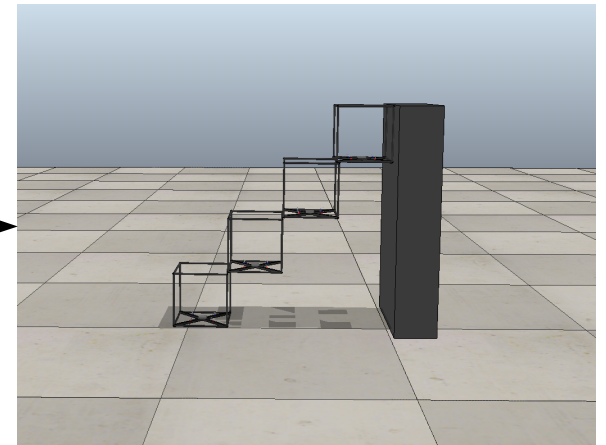
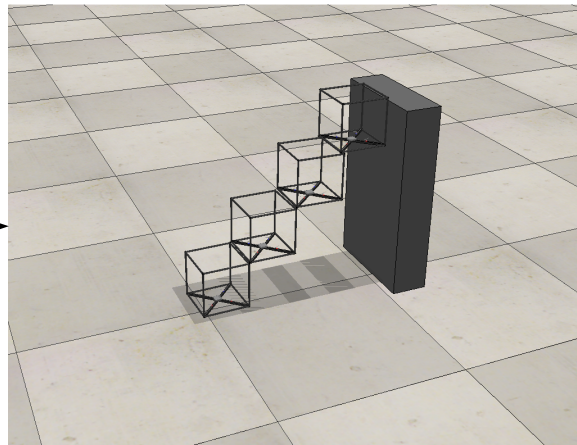
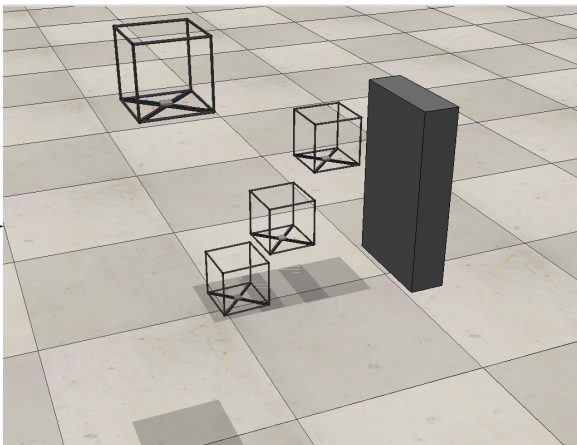
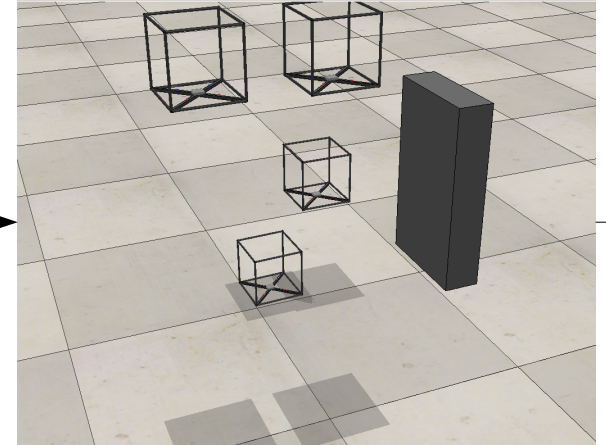
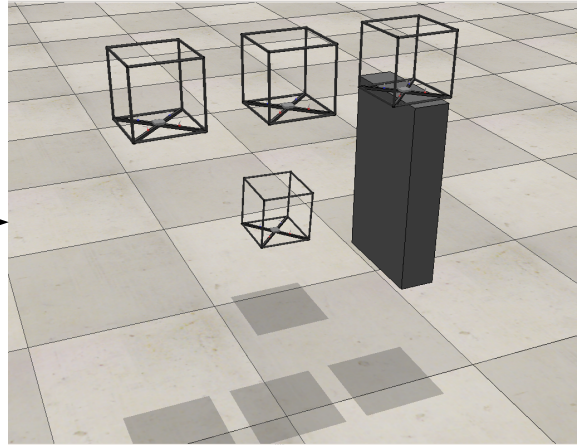
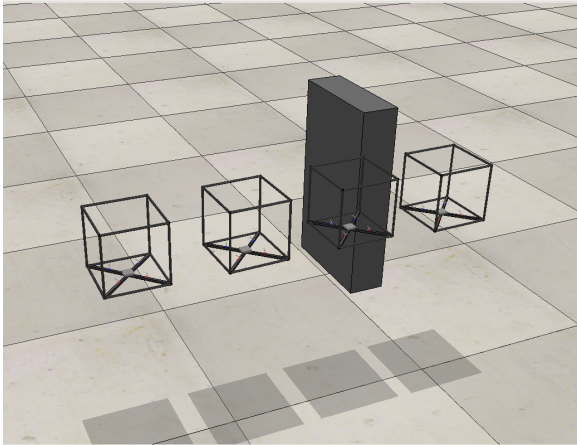
- Multiple quadrotors used to build a structure
- Centralized planning
- Non-scalable sensing
- No learning applied
- Classical PID controllers



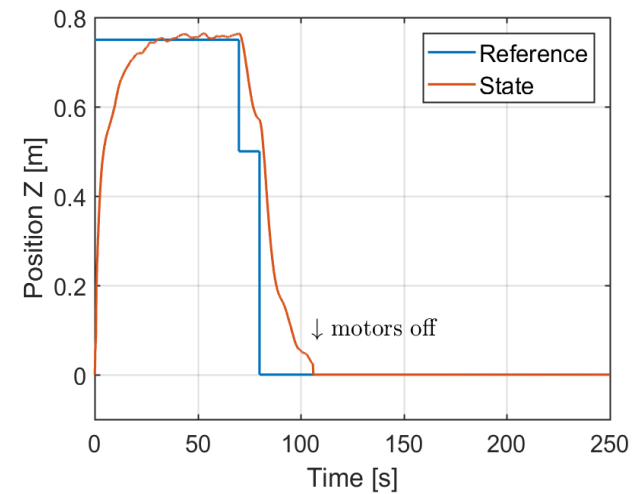
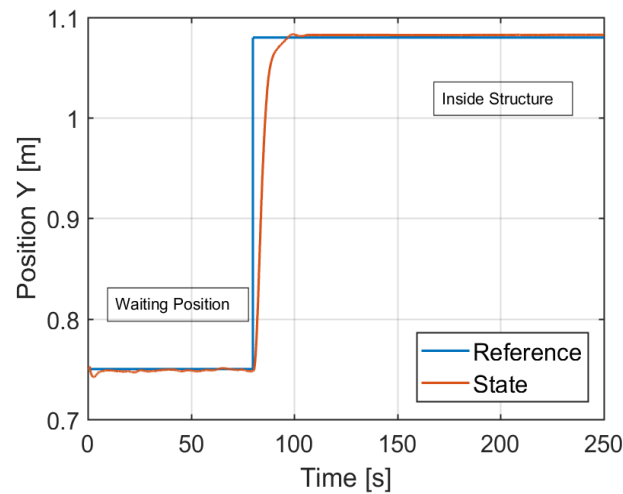
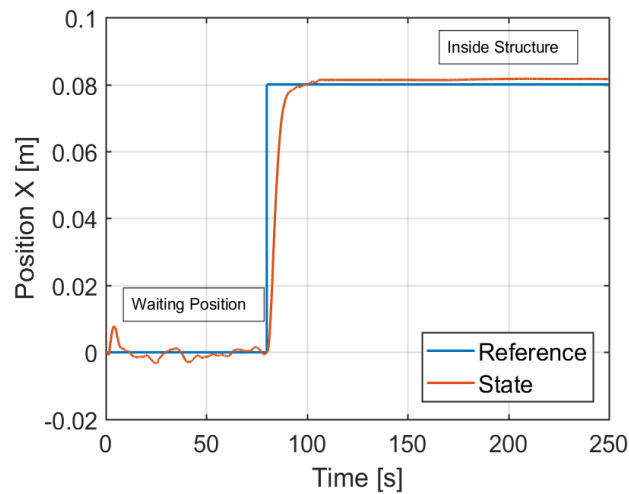
Pyramidal Structure



Stairs



Position of one of the vehicles





General Comments

- The architecture shows the flow of information
 - Independent on the type of structure
 - Independent on the type of robots
- Why do we show two structures?
 - Pyramid - Planning
 - Stairs - Control



Conclusion

- A high-level architecture is proposed for systems where the building blocks are the robots
- The architecture works for multiple-vehicle scenarios
- We used the architecture to build two structures in simulation, using simplified approach for each one of the System Modules